Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 9.1 · Transforming

For every real number x, consider the expressions

$$A = e^{x}e^{2-x}, \qquad B = \frac{(e^{2x})^{3}}{e^{x}}, \qquad C = e^{-x}(e^{x}+1).$$

1. Simplify A and B, justifying each grouping of exponents.
2. Expand then simplify C. Explain why a sum does not simplify in the same way as a product.

Exercise 9.2 · Equations and inequalities

Consider an equation and an inequality in the real unknown x:

$$e^{2x-1} = e^{x+3} \qquad \text{and} \qquad e^{-3x} \geq e^{x-2}.$$

1. Solve each in $\mathbb{R}$, citing the property of the exponential that allows the exponents to be compared.
2. While solving the inequality, explain at which step the direction of the inequality changes. Give the solution set as an interval.

Exercise 9.3 · Product and maximum

The function f is defined on $[0,4]$ by $f(x) = xe^{-x}$.

1. Calculate $f\prime(x)$, then study its sign on the interval.
2. Draw the monotonicity table of f. Deduce its maximum and the x-coordinate where it is reached.
3. Check that the values at the endpoints do not give a larger maximum. Keep the exact values.

Exercise 9.4 · A modelled population

The following two parts are independent.

Part A: A population. A population is modelled by $P(t) = 200e^{0.1t}$, where $t \geq 0$ is the time elapsed in years. $P(t)$ is a theoretical count, which is not necessarily an integer.

1. Determine the initial count, then the multiplicative factor between times t and $t+1$. Deduce the exact annual rate and express it as a percentage rounded to two decimal places.
2. Explain why this model cannot necessarily describe a real population indefinitely.
3. **Part B: Two curves.** In the same coordinate system, sketch the curves of $t \mapsto e^{\frac{t}{2}}$ and $t \mapsto e^{-\frac{t}{2}}$. State where they increase or decrease, their common point and their symmetry. Identify each curve clearly.

Exercise 9.5 · Searching for a threshold

The quantity of a substance, in milligrams, is modelled at time $t \geq 0$ by $Q(t) = 80e^{-\frac{t}{2}}$, where t is expressed in hours. Checks take place at integer times $n = 0, 1, 2, \ldots$. We look for the first check at which the quantity is strictly less than 10 mg.

1. Write an algorithm that searches for this first integer. Provide an iteration bound and state what the program returns if the bound is reached.
2. Run the algorithm, then check numerically the quantity at the index found and at the previous index.
3. Justify, using the decrease of Q, that this double check establishes that it is indeed the first index. Do not use a logarithm.

Exercise 9.6 · A symmetry

Consider the function f defined on $\mathbb{R}$ by $f(x) = e^x + e^{-x}$.

1. Study the parity of f and interpret the result for its curve.
2. Prove that $f(x) \geq 2$ for every real number x, choosing a method. State the equality case.
3. Suggest a check of the result using a second method, based on a square or a study of monotonicity.

Exercise 9.7 · Uniqueness under a hypothesis

Extension. Assume that the exponential function has already been constructed. Let f be a function differentiable on $\mathbb{R}$ that satisfies $f\prime(x) = f(x)$ for every real number x and $f(0) = 1$. Let $g(x) = f(x)e^{-x}$.

1. Calculate $g\prime$ and explain why g is constant on $\mathbb{R}$.
2. Determine this constant, then deduce the expression of f.
3. Does this proof establish the existence or the uniqueness of the function sought? Justify.

Exercise 9.8 · Delayed review

Define the sequence (u_n) by $u_n = e^{2n+1}$ for every natural number n.

1. Prove that this sequence is geometric, then state its first term and common ratio.
2. Calculate $u_0 + u_1 + u_2$ in exact form. Compare the direct calculation with the use of a sum formula.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 18 p. 187, questions a, b and c. Simplify the products and quotients, citing the property of exponentials used.

CQFD, exercise 30 p. 187, questions a, b and c. Solve the three inequalities and write the solution sets as intervals.